
Problem 8.21. For any positive integer x, let x^R be the integer whose binary representation is the reverse of the binary representation of x (Assume no leading 0s in the binary representation of x.) Define the function $R^+ : \mathbb{N} \longrightarrow \mathbb{N}$ where $R^+(x) = x + x^R$.

Part a. Let $A_2 = \{\langle x, y\rangle \mid R^+(x) = y\}$. Show $A_2 \in \mathrm{L}$.

Proof. To show that $A_2 \in \mathrm{L}$, we give a log space algorithm M that decides A_2.

$M =$ "On input $\langle x, y\rangle$, where x and y are positive binary integers without leading 0s:

1. Let i, j, n_x, n_y be binary integers, and s and c be binary digits on work tape.
2. Let $n_x = |x|$, and $n_y = |y|$.
3. Let $i = 1$, $j = n_x$, and $c = 0$.
4. Repeat for $n_x - 1$ steps:
5. If y does not have i^{th} digit, then *reject*.
6. Let s be the sum of ith and j^{th} binary digits of x and c. If the addition produces a carry, then $c = 1$. Otherwise, $c = 0$.
7. If s and i^{th} digit of y are different, then *reject*.
8. Increment i by 1, and decrement j by 1.
9. If $c = 1$, then:
10. If y does not have i^{th} digit, then *reject*.
11. If c and i^{th} digit of y are different, then *reject*.
12. If y has $(i + 1)^{th}$ digit, then *reject*.
13. *Accept*."

□

Part b. Let $A_3 = \{\langle x, y\rangle \mid R^+(R^+(x)) = y\}$. Show $A_3 \in \mathrm{L}$.

Proof. □